

Modeling and Control of Salient-Pole Permanent Magnet Axial-Gap Self-Bearing Motor

Quang Dich Nguyen, *Student Member, IEEE*, and Satoshi Ueno, *Member, IEEE*

Abstract—This paper will introduce an inset-type axial-gap self-bearing motor (AGBM), which is an electrical combination of an axial thrust-bearing and an axial flux motor. A method for controlling the axial position and the rotating speed of the AGBM without the influence of reluctance torque is also discussed. First, the axial force and the motoring torque are analyzed theoretically, and then, the control method is derived. To demonstrate the proposed technique, an inset two-pole AGBM has been constructed and tested. The experimental results confirm that the motor works stably with the proposed vector control. Moreover, the rotating torque and the axial force can be controlled independently as well.

Index Terms—Axial-gap self-bearing motor (AGBM), bearingless motor drives, combined bearing motor, salient-pole permanent magnet synchronous motor (PMSM), self-bearing motor.

I. INTRODUCTION

RECENTLY, active magnetic-bearing motors have been designed to overcome the limitations of the conventional mechanical-bearing motors. Magnetic-bearing motors can work in all environments without lubrication and do not cause contamination; further, they can run at very high speeds. Therefore, they are very valuable machines with a number of novel features, and with a vast range of diverse applications [1].

The conventional magnetic-bearing motor usually has a rotary motor installed between two radial magnetic bearings, or a mechanical combination of a rotary motor and a radial magnetic bearing, as shown in Figs. 1 and 2 [2]–[4]. The radial magnetic bearings create radial levitation forces for rotor, while an axial magnetic bearing produces a thrust force to keep the rotor in the correct axial position relative to the stator.

However, these magnetic-bearing motors are large, heavy, and complex in control and structure, which cause problems in applications that have limited space. Thus, a simpler and smaller construction and a less complex control system are desirable.

An electrical combination of radial bearing and motor was introduced in [5]. The proposed bearingless brushless dc motor consists of five concentrated coils, which generate radial forces and motor torque simultaneously. The tilting deflection and the axial position of the rotor are stabilized passively. In this, motor force and torque generation are coupled and highly nonlinear, thus, a complex control method is required to achieve high operation performance.

Manuscript received September 23, 2009; revised January 3, 2010; accepted February 26, 2010. Date of publication April 8, 2010; date of current version May 6, 2011. Recommended by Technical Editor J. Xu. This work was supported in part by the Japanese Ministry of Education, Science, and Culture.

The authors are with the Control Engineering Laboratory, Department of Mechanical Engineering, Ritsumeikan University, Kusatsu 525-8577, Japan (e-mail: gr046070@ed.ritsumeikan.ac.jp).

Digital Object Identifier 10.1109/TMECH.2010.2045392

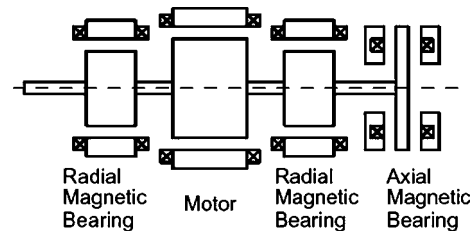


Fig. 1. Structure of conventional magnetic-bearing motor.

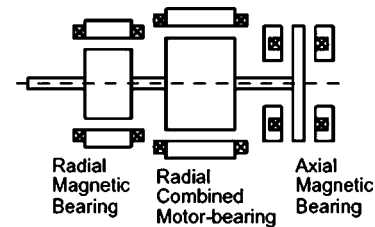


Fig. 2. Structure of radial-combined magnetic-bearing motor.

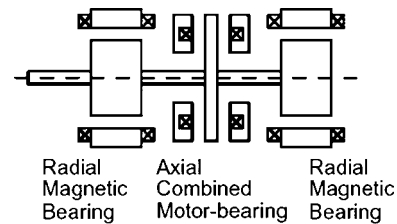


Fig. 3. Structure of axial-gap self-bearing motor.

An axial magnetic bearing is composed of a rotary disc fixed on a rotary shaft and electromagnets arranged on both sides of the disc at a proper minute distance. This structure is similar to that of an axial-flux ac motor. Therefore, an axial-gap bearingless motor has been introduced, as shown in Fig. 3 [6]–[14].

The axial-combined motor-bearing in [6] and [7] has four-pole motor windings and two-pole suspension windings, it is hence mechanical combination of a disc motor and an axial magnetic bearing. Further, in this motor, the axial force is produced from both d - and q -axis currents. The structure and control are still complicated.

Ueno and Okada have introduced an electrically combined motor-bearing in [8] and [9], in which the stator has only three-phase windings; however, the motor can simultaneously provide noncontact levitation and rotation. This motor is then called an axial-gap self-bearing motor (AGBM). Obviously, it is simpler in structure and control, since hardware components can

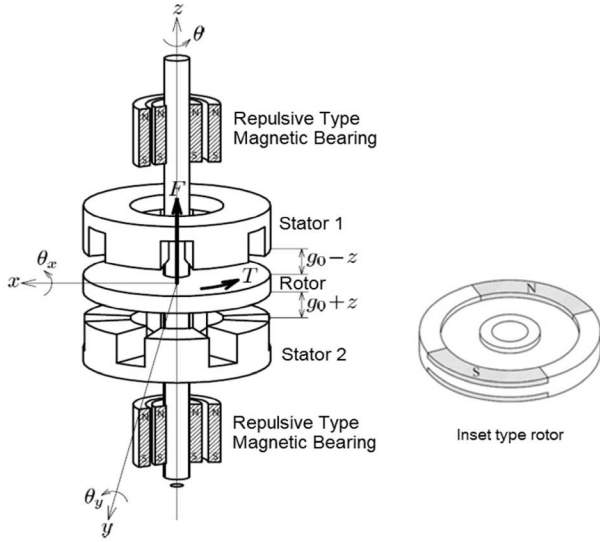


Fig. 4. Detailed structure of the AGBM.

be reduced. The AGBM can be realized as an induction motor (IM) [8], or a permanent magnet (PM) motor [9]–[14]. The PM motor is given special attention, because of its high-power factor, high efficiency, and simplicity in production. The control method for them was developed under the vector control principle, where the direct-axis current i_d is used for controlling the axial force and the quadrature-axis current i_q is used for controlling the rotating torque. The result showed that the motor has capability of producing torque and axial force simultaneously.

Nowadays, most of researches dealt only with the surface-mounted nonsalient PM AGBM [9]–[14]. This kind of AGBM was considered having more simple control system, because of the equal inductances along both d and q axes. However, it has large air gap between the rotor and the stator, low torque, and levitation force; whereas, the inset-type PM AGBM, featured by small air gap, high torque, and suspension force, is not much mentioned. The difficulty with the inset salient-pole rotor stays in reluctance torque, caused by the difference between the d - and q -axis inductances, which strongly affects the quality of the axial position and speed control.

In this paper, the model of the salient two-pole AGBM with double stators is introduced and analyzed. A closed-loop vector control method for the axial position and the speed is developed in the way of eliminating the influence of the reluctance torque. Moreover, the compensational method for reference currents based on the difference between d - and q -axis inductances is also recommended. The proposed control method can be utilized by any type of PM AGBM, such as a nonsalient AGBM ($L_d = L_q$) and salient AGBM ($L_d < L_q$). In order to confirm the presented technique, an experimental setup has been produced and tested.

II. MATHEMATICAL MODEL

Fig. 4 illustrates the structure of the proposed axial-gap self-bearing motor. The radial motions x , y , θ_x , and θ_y of the rotor are constrained by radial magnetic bearings, such as the repulsive

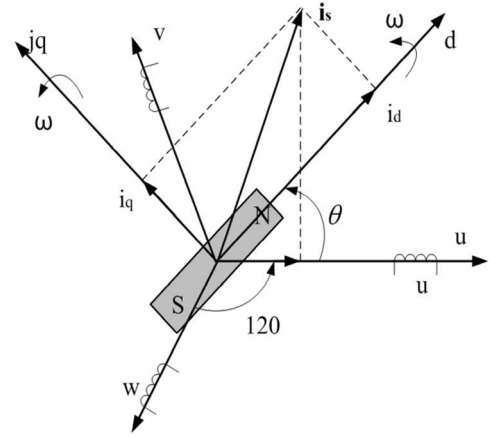


Fig. 5. Coordinates.

bearing. Only rotational motion and translation of rotor along the z -axis are considered. The motor has 2 DOF.

The rotor is a flat disc with PMs inserted on two faces of the disc to create a salient-pole rotor. Two stators, one in each rotor side, have three-phase windings that generate rotating magnetic fluxes in the air gap. These produce the motoring torque T_1 and T_2 on the rotor and generate the attractive force F_1 and F_2 between the rotor and the stators. The total motoring torque T is the sum of these torques, and the axial force F is the difference of the two attractive forces.

To obtain mathematical model of the AGBM, the axial force F_s and motoring torque T_s are first calculated for one stator. Similar to the conventional PM motor, the mathematical model of the AGBM is presented in a rotor-field-oriented reference frame using the so-called d , q coordinates, which is indicated in Fig. 5. The d -axis is aligned with the PM flux vector. The axes u , v , and w indicate the direction of the flux produced by the corresponding phase windings. The phase difference between the u -axis and d -axis is angular position θ of the rotor or the rotor flux vector. The power invariant principle is used for transforming between coordinates. Let a represent a stator quantity to be transformed (current, voltage, or flux), the transformation in matrix form is written as follows:

$$\begin{bmatrix} a_d \\ a_q \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos \theta & \cos(\theta - 120^\circ) & \cos(\theta + 120^\circ) \\ \sin \theta & \sin(\theta - 120^\circ) & \sin(\theta + 120^\circ) \end{bmatrix} \begin{bmatrix} a_u \\ a_v \\ a_w \end{bmatrix}. \quad (1)$$

Since the PM with unity permeability is used, the rotor is salience; therefore, the self-phase inductance of the stator is dependent on the rotor angular position, which means that the d -axis inductance is different from q -axis inductance. Furthermore, the self-phase inductance is a function of the air gap g between the rotor and the stator. Normally, the self-phase inductance is inversely proportional to the air gap; therefore, the d - and q -axis phase inductance of the stator windings may be

approximated by

$$\begin{cases} L_{sd} = \frac{3L'_{sd0}}{2g} + L_{sl} \\ L_{sq} = \frac{3L'_{sq0}}{2g} + L_{sl} \end{cases} \quad (2)$$

where L'_{sd0} and L'_{sq0} are the effective inductances per unit gap in d and q axes, respectively, and L_{sl} is leakage inductance.

Then, the stator voltage and the flux of the AGBM in the d, q coordinates can be expressed in the following equations:

$$\begin{cases} u_{sd} = R_s i_{sd} + L_{sd} \frac{di_{sd}}{dt} - \omega L_{sq} i_{sq} \\ u_{sq} = R_s i_{sq} + L_{sq} \frac{di_{sq}}{dt} + \omega L_{sd} i_{sd} + \omega \lambda_m \\ \lambda_{sd} = L_{sd} i_{sd} + \lambda_m \\ \lambda_{sq} = L_{sq} i_{sq} \end{cases} \quad (3)$$

where λ_m is the flux linkage caused by PM. For simplicity, the magnetic flux of the rotor is replaced by an equivalent winding with current i_f and inductance L_f . The rotor flux can be expressed only in d -axis as follows:

$$\lambda_f = \lambda_{fd} = i_f L_f + L_m i_{sd} \quad (4)$$

with

$$L_f = \frac{3}{2} \frac{L'_{sd0}}{g} + L_{fl} \quad (5)$$

and mutual inductance

$$L_m = \frac{3L'_{sd0}}{2g}. \quad (6)$$

From (2) to (4), the magnetic energy in the motor is calculated as follows:

$$W = \frac{\lambda_f i_f + \lambda_{sd} i_{sd} + \lambda_{sq} i_{sq}}{2}. \quad (7)$$

Therefore, the attractive force of one stator is received by the derivative of the magnetic energy with respect to the air gap

$$F_s = -\frac{\partial W}{\partial g} = \frac{3L'_{sd0}}{4g^2} (i_{sd} + i_f)^2 + \frac{3L'_{sq0}}{4g^2} i_{sq}^2 \quad (8)$$

and the motoring torque for one stator is calculated as follows:

$$\begin{aligned} T_s &= P(-\lambda_{sd} i_{sq} + \lambda_{sq} i_{sd}) \\ &= \frac{3PL'_{sd0}}{2g} i_f i_{sq} + \frac{3P(L'_{sd0} - L'_{sq0})}{2g} i_{sd} i_{sq} \end{aligned} \quad (9)$$

where P is the number of pole pairs.

From (9), the output torque is a combination of an excitation torque and a reluctance torque. This means, in every operation mode, the motor has to produce an additional torque to compensate the reluctance torque. In the nonsalient pole rotor, this reluctance torque can be ignored to make control system more simple. However, in the salient-pole rotor when the reluctance torque can reach the relative high amplitude, neglecting this torque component will reduce the quality of system, especially in operation mode with axial load ($i_d \neq 0$).

From (8) and (9), F_1 and T_1 are calculated by substituting $g = g_0 - z$, $i_{sd} = i_{d1}$, and $i_{sq} = i_{q1}$, and F_2 and T_2 are calculated by substituting $g = g_0 + z$, $i_{sd} = i_{d2}$, and $i_{sq} = i_{q2}$. Thus, the total axial force F and torque T are given by

$$F = F_2 - F_1 \quad (10)$$

$$T = T_1 + T_2 \quad (11)$$

where g_0 is the axial gap at the equilibrium point, and z is the displacement.

For linearization at the equilibrium point ($z = 0$), (10) and (11) are expanded into a Maclaurin series and the first-order term is taken, to yield

$$\begin{aligned} T &= K_T (i_{q1} + i_{q2}) + K_T (i_{q2} - i_{q1}) \frac{z}{g_0} \\ &\quad + K_R (i_{d1} i_{q1} + i_{d2} i_{q2}) + K_R (i_{d2} i_{q2} - i_{d1} i_{q1}) \frac{z}{g_0} \end{aligned} \quad (12)$$

$$\begin{aligned} F &= K_{Fd} \{(i_{d2} + i_f)^2 - (i_{d1} + i_f)^2\} + K_{Fq} (i_{q2}^2 - i_{q1}^2) \\ &\quad + 2K_{Fd} \{(i_{d2} + i_f)^2 + (i_{d1} + i_f)^2\} \frac{z}{g_0} \\ &\quad + 2K_{Fq} (i_{q2}^2 + i_{q1}^2) \frac{z}{g_0} \end{aligned} \quad (13)$$

where $K_{Fd} = 3L'_{sd0}/4g_0^2$ and $K_{Fq} = 3L'_{sq0}/4g_0^2$ are the force factors, and $K_T = -3PL'_{sd0}i_f/2g_0$ and $K_R = -3(L'_{sd0} - L'_{sq0})/2g_0$ are the torque factors.

III. CONTROL PRINCIPLE

To increase the total moment twice, the component moment created by one stator, the moment-generated currents for both stators must be of same direction and value. To keep the rotor in right position between two stators, the forces acting on rotor from both sides must be of same value but inverse, i.e., under the effect of the axial load, if the force-generated current of one side increases, then correspondingly, that current of other side has to decrease the same amount. The rotating torque can be controlled effectively by using the quadrature axis current, and the axial force can be controlled by changing the direct-axis current. It is supposed that

$$\begin{cases} i_{q1} = i_{q2} = i_q \\ i_{d1} = i_{d0} - i_d \\ i_{d2} = i_{d0} + i_d \end{cases} \quad (14)$$

where i_{d0} is an offset current, and the value can be zero or a small value around zero. Substituting (14) into (12) and (13) gives

$$\begin{aligned} T &= 2K_T i_q + 2K_R i_{d0} i_q + \frac{2K_R i_d i_q z}{g_0} \\ &= T_{\text{eff}} + T_{rl0} + T_{rlz} \end{aligned} \quad (15)$$

$$\begin{aligned} F &= 4K_{Fd} (i_f + i_{d0}) i_d + 4\{K_{Fd} (i_d^2 + i_{d0}^2 + i_f^2) \\ &\quad + 2K_{Fd} i_f i_{d0} + K_{Fq} i_q^2\} \frac{z}{g_0}. \end{aligned} \quad (16)$$

From (15), the total torque consists of the following three components.

- 1) The first component, $T_{\text{eff}} = 2K_T i_q$, is the efficient torque; this is main component of the output torque.
- 2) The second one, $T_{r10} = 2K_R i_{d0} i_q$, is the reluctance torque caused by current i_{d0} . Therefore, assuming that $i_{d1} = -i_{d2} = -i_d$, i.e., $i_{d0} = 0$, then this reluctance torque is eliminated.
- 3) The last one, $T_{rlz} = 2K_R i_d i_q z / g_0$, is reluctance torque caused by current i_d under the effect of the displacement z . When the displacement is well controlled to be zero, or very small in comparison with air gap at the equilibrium point g_0 , the influence of this component can be neglected. Then, the total torque becomes as follows:

$$T = 2K_T i_q. \quad (17)$$

Using aforementioned control law, the total axial force is received as follows:

$$F = 4K_{Fd} i_f i_d. \quad (18)$$

Obviously, the effect of the inductance difference to axial force is also vanished.

From (17) and (18), it is easy to see that the total torque can be controlled with the quadrature-axis current and the axial force can be controlled with the direct-axis current. And in combination with (3), the mathematical model of the AGBM is totally constructed with voltage, force, and torque equations. They are simple linear equations; therefore, the control system can be easily implemented with the conventional controllers.

A. Axial Displacement Control

For simplicity, it is assumed that the radial motion of the rotor is restricted by the ideal radial bearings. Therefore, the axial motion of the rotor is independent of the radial motion. The dynamic equation of the axial motion of the rotor is

$$F = m\ddot{z} \quad (19)$$

where m is the mass of the moving parts, and F is the axial force shown in (16). Substituting (16) into (19) with $i_{d0} = 0$ gives

$$m\ddot{z} = 4K_{Fd} i_f i_d + \{4K_{Fd}(i_d^2 + i_f^2) + 4K_{Fq} i_q^2\} \frac{z}{g_0}. \quad (20)$$

This can be summarized as follows:

$$m\ddot{z} + K_z z = K_m i_d \quad (21)$$

where $K_z = -4K_{Fd}(i_d^2 + i_f^2) - 4K_{Fq} i_q^2$ is the stiffness of the motor, and $K_m = 4K_{Fd} i_f$ is the force gain. It is easy to realize that K_z is negative, which means that this system is unstable. To stabilize the system, a controller with the derivative component must be used. Assuming that the proportional-derivative (PD) controller is used, the output of the controller will represent the direct-axis reference current, i.e.,

$$i_d = -K_p z - K_d \dot{z} \quad (22)$$

where K_p is the proportional constant, and K_d is the derivative constant of the axial-position controller. Substituting (22) into

(21) gives

$$m\ddot{z} + K_m K_d \dot{z} + (K_z + K_m K_p) z = 0. \quad (23)$$

The system becomes stable only when all constant coefficients of the polynomial function have the same sign. Therefore, when $K_d > 0$, the system will be stable if the proportional constant satisfies the following condition:

$$K_p > -\frac{K_z}{K_m} = \frac{K_{Fd}(i_d^2 + i_f^2) + K_{Fq} i_q^2}{K_{Fd} i_f}. \quad (24)$$

Actually, steady-state error occurs when only the PD controller is used, and to remove this, a PID controller should be used.

B. Speed Control

It is assumed that the axial position can be controlled well, so that the displacement becomes zero or very small in comparison with the air gap at the equilibrium point. The dynamic equation of the rotating motion of the rotor is as follows:

$$T = T_L + J\dot{\omega} + B\omega \quad (25)$$

where T_L is the load moment, J is the inertia moment of whole system, and B is the damping factor.

Obviously, the system is stable; so therefore, the design of the speed controller is similar to the procedure used for a dc motor.

IV. CONTROL STRUCTURE

As stated earlier, the motoring torque of the AGBM can be controlled by the q -axis current (i_q), while the axial force can be controlled by the d -axis current (i_d). Fig. 6 shows the control scheme proposed for the AGBM drive, with the decoupled d - and q -axis current controller.

The axial displacement from the equilibrium point along the z -axis z can be detected by the gap sensor. The detected position is compared with the axial-position command z_{ref} , and the difference is input to the axial-position controller R_z . The output of the axial-position controller is used to calculate the d -axis reference current with compensation procedure from (16). The position command z_{ref} is always set to zero to ensure that the rotor is at the midpoint between two stators. The d -axis reference currents for two stator windings $i_{d1\text{ref}}$ and $i_{d2\text{ref}}$ can be generated by using the offset current i_{d0} , and respectively, subtracting or adding $i_{d\text{ref}}$. In this paper, the value of the offset current is zero.

The rotor speed detected from encoder is compared with the reference speed, and the difference is input to the speed controller R_ω . The output of the speed controller is used to calculate the q -axis reference current by using (15). The q -axis reference currents for two stator windings are same with this current.

The motor currents in the two-phase stator reference frame α , β are calculated by measuring two actual phase currents. Therefore, the d - and q -axis components are obtained using the rotor position from the encoder. The quadrature components are controlled by the reference value that is given by the speed

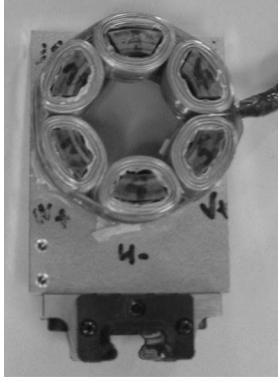


Fig. 9. Stator of the AGBM.

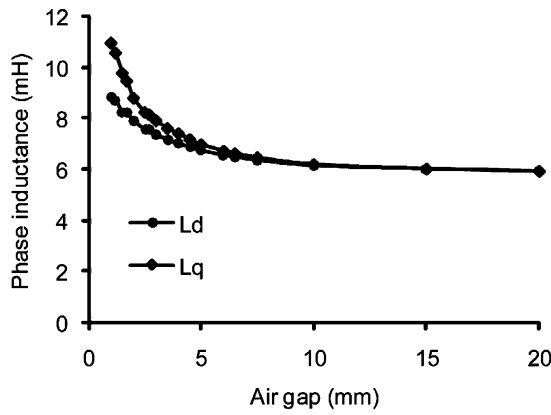


Fig. 10. Static phase inductances of the AGBM.

(DAC)] and two encoder interfaces. The DSP reads the displacement signal from the displacement sensor via an ADC, and the rotor angle position and speed from the encoder via an encoder interface. Two motor phase currents are sensed, rescaled, and converted to digital values via an ADC. The dSpace1104 then calculates reference currents using the rotation control and axial-position control algorithms and send its commands to the three-phase inverter board. The AGBM is supplied by the two three-phase PWM inverters, with a switching frequency of 20 kHz.

B. Experimental Results

First, parameter estimation is carried out. The phase resistor is 2.6Ω . The static phase inductances are measured by measuring the frequency response of the voltage versus the current of the stator coil. The results are shown in Fig. 10. This gives $L'_{sq0} = 9.6 \times 10^{-6}$ Hm, $L'_{sd0} = 8.2 \times 10^{-6}$ Hm, and $L_{sl} = 6 \times 10^{-3}$ H. The air gap between the stator and the rotor is adjusted to 1.7 mm. The rotor mass is 0.235 kg, and the rotor inertia is $0.000086 \text{ kg}\cdot\text{m}^2$. The amplitude of the flux induced by the PMs of the rotor in the stator phases can be estimated by measuring no-load voltage of the AGBM in generator mode. The result is shown in Fig. 11, which gives $\lambda_m = 0.0126$ Wb.

To find the relation between the axial force and the direct-axis current, the experimental setup is placed upright, and the

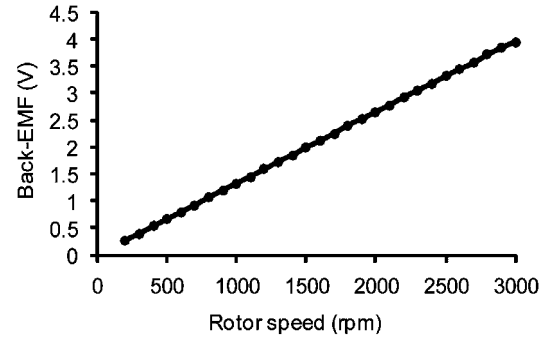


Fig. 11. Back electromotive force (EMF) of the AGBM.

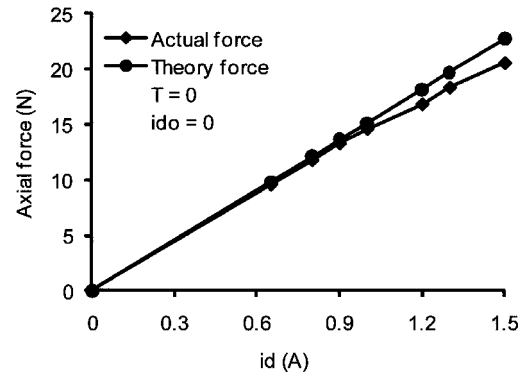


Fig. 12. Actual and theoretical axial force.

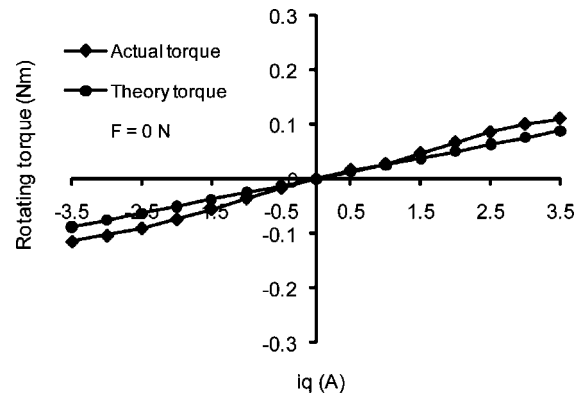


Fig. 13. Actual and theoretical torque.

stators bear the axial load on its surface, while the motor is at a standstill. The current i_d is measured for different values of the axial load, and the result is illustrated in Fig. 12. Its linear shape confirms the proposed theory. At high axial load, the actual force is lower than the theoretical force because of an increase in the demagnetization process. A reduction in the magnetic flux will cause a decline in the axial force.

Fig. 13 shows the relation between the total torque and the quadrature-axis current when the axial load is zero. By supplying various values of the current i_q , the static torque is measured using load cell. The theoretical excitation torque is calculated from (16) and is smaller than actual torque, which means the motor produces an extra reluctance torque. In static mode, the effect of the reluctance torque in both sides of the AGBM cannot

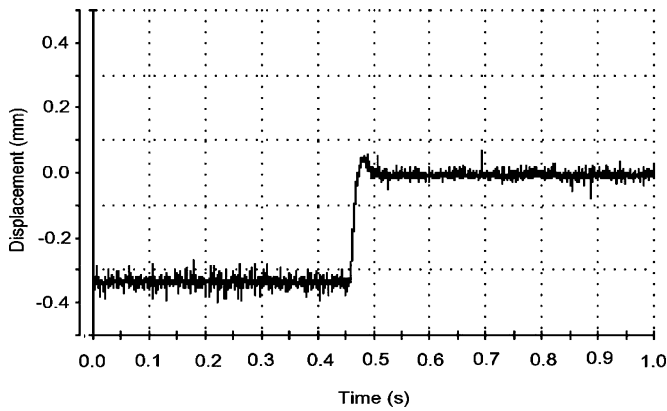


Fig. 14. Response of displacement at zero speed.

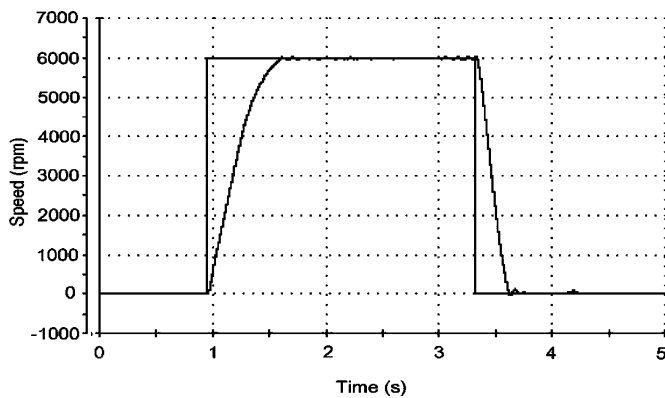


Fig. 15. Response of speed at start.

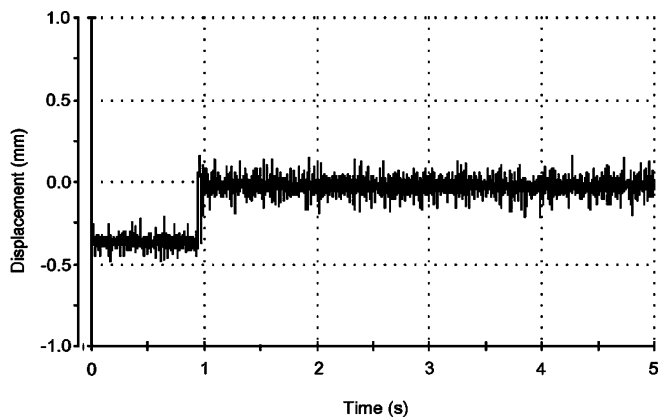


Fig. 16. Response of displacement and speed at start.

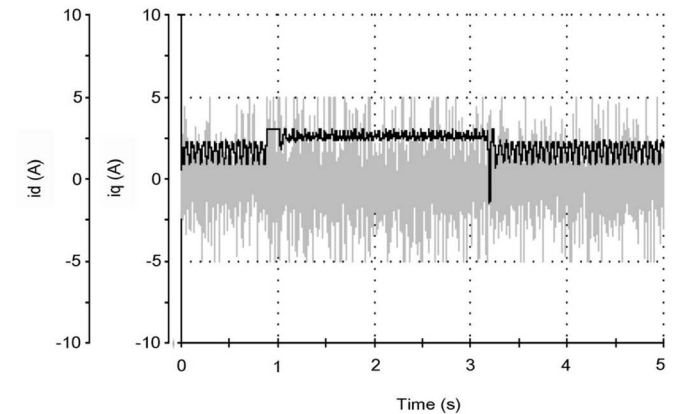
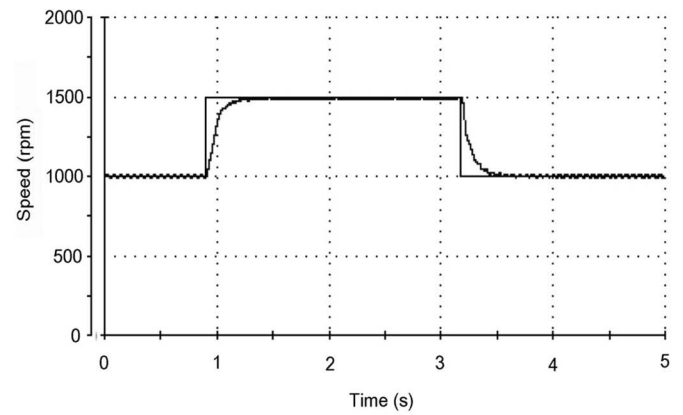
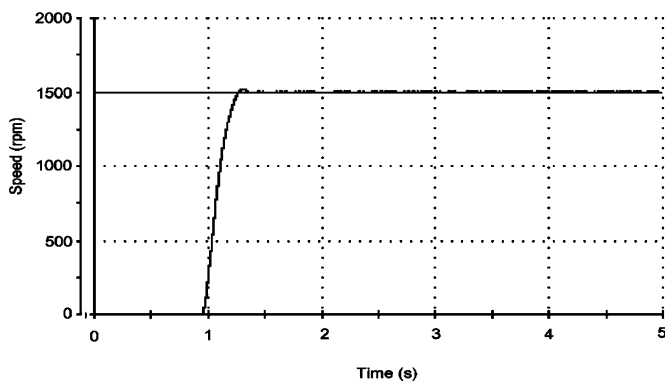


Fig. 17. Currents and displacement when rotor speed was changed.

be removed because the current i_d is not controlled to satisfy (14). These reluctance torques then make the totally static torque of the motor to become higher than theoretical excitation torque.

Fig. 14 shows the axial displacement at 0 r/min. The original displacement is set to 0.32 mm, and at the time of 0.45 s, the axial-position controller starts to work. In transient state, the maximum error is 0.05 mm, much smaller than the air gap at the equilibrium point ($g_0 = 1.7$ mm) and the settling time is about 0.05 s. After that, the displacement is almost zero in a steady state.

Fig. 15 describes the change in the speed from 0 to 6000 r/min and *vice versa* when the displacement is zero and the limited current is ± 5 A. With small starting time (about 0.7 s) and

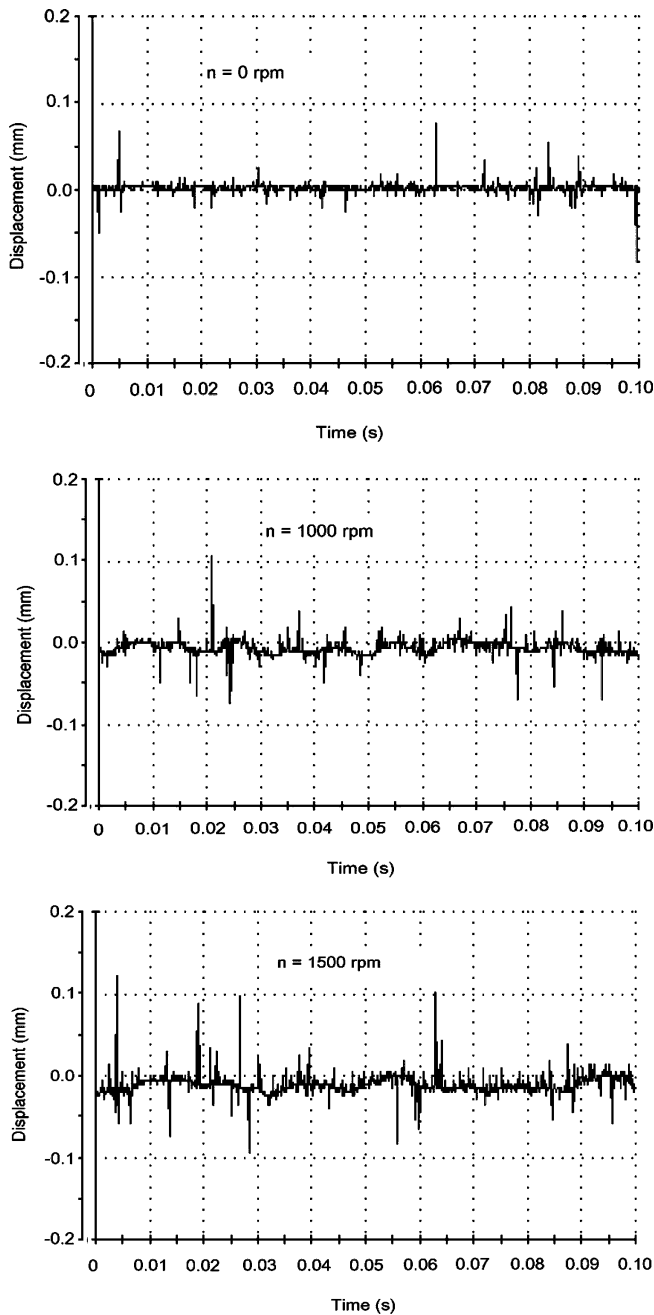


Fig. 18. Displacement at different speeds.

stopping time (about 0.4 s), the motor drive shows its good dynamic response.

Fig. 16 shows the displacement and the speed when motor starts to work. The displacement jumps immediately to zero and the rotor speed reaches 1500 r/min after 0.5 s without influence of each other.

Obviously, the displacement controller and speed controller work independently with each other.

Fig. 17 illustrates the change of the direct-axis current i_d , the quadrature-axis current i_q , and the displacement when the motor speed changes from 1000 to 1500 r/min and *vice versa*. Fig. 18 shows more detail about the displacement at different speeds. It

can be seen that at the higher speed, there is a little vibration in the displacement, but the maximum error at 1500 r/min is about 0.025 mm, much smaller than the air gap. One of the reasons that causes this vibration is mechanical problem that comes of the long rotor shaft. The fluctuation of the rotor can occur at a high rotational speed because of the long shaft. Other reason comes from the effect of the quadrature axis current.

VI. CONCLUSION

The AGBM was fabricated with an inset PM-type rotor and vector control was implemented. The results confirm that the motor can perform both functions of motor and axial bearing without any additional windings. The reluctance torque and its influence are rejected entirely. Although, there is very little interference between the axial-position control and speed control in all operating mode, the proposed AGBM can be used for many kind of applications, which require small air gap, high speed, and levitation force, such as liquid pumps, compressors, and machine tools.

REFERENCES

- [1] M. Dussaux, "The industrial application of the active magnetic bearing technology," presented at the 2nd Int. Symp. Magn. Bearings, Tokyo, Japan, Jul. 12–14, 1990.
- [2] Y. Okada, K. Dejima, and T. Ohishi, "Analysis and comparison of PM synchronous motor and induction motor type magnetic bearing," *IEEE Trans. Ind. Appl.*, vol. 32, no. 5, pp. 1047–1053, Sep./Oct. 1995.
- [3] Z. Ren and L. S. Stephens, "Closed-loop performance of a six degree-of-freedom precision magnetic actuator," *IEEE/ASME Trans. Mechatronics*, vol. 10, no. 6, pp. 666–674, Dec. 2005.
- [4] T. Schneider and A. Binder, "Design and evaluation of a 60 000 rpm permanent magnet bearingless high speed motor," in *Proc. Int. Conf. Power Electron. Drive Syst.*, Nov. 2007, pp. 1–8.
- [5] H. Grabner, W. Amrhein, S. Silber, and W. Gruber, "Nonlinear feedback control of a bearingless brushless DC motor," *IEEE/ASME Trans. Mechatronics*, vol. 15, no. 1, pp. 40–47, Feb. 2010.
- [6] O. Ichikawa, T. Fukao, and A. Chiba, "Construction method of a position controller of axial gap bearingless motor," (in Japanese), in *Proc. IEE Jpn. Tech. Meeting Rotating Machinery*, 1998, vol. 120, pp. 127–132, Paper RM-98-141.
- [7] F. Saito, O. Ichikawa, K. Takahashi, and T. Fukao, "Hierarchical levitation and rotation control system of axial-gap bearingless motors," (in Japanese), in *Proc. IEE Jpn. Tech. Meeting Semicond. Power Converter*, 1999, pp. 61–66, Paper SPC-99-11.
- [8] S. Ueno and Y. Okada, "Vector control of an induction type axial gap combined motor-bearing," in *Proc. IEEE Int. Conf. Adv. Intell. Mechatronics*, Atlanta, GA, Sep. 19–23, 1999, pp. 794–799.
- [9] S. Ueno and Y. Okada, "Characteristics and control of a bidirectional axial gap combined motor-bearing," *IEEE Trans. Mechatronics*, vol. 5, no. 3, pp. 310–318, Sep. 2000.
- [10] M. Horz, H.-G. Herzog, and N. Medler, "System design and comparison of calculated and measured performance of a bearingless BLDC-drive with axial flux path for an implantable blood pump," in *Proc. Int. Symp. Power Electron., Electr. Drives, Autom. Motion (SPEEDAM 2006)*, May, pp. 1024–1027.
- [11] Y. Okada, N. Yamashiro, K. Ohmori, T. Masuzawa, T. Yamane, Y. Konishi, and S. Ueno, "Mixed flow artificial heart pump with axial self-bearing motor," *IEEE/ASME Trans. Mechatronics*, vol. 10, no. 6, pp. 658–665, Dec. 2005.
- [12] D. Q. Nguyen and S. Ueno, "A study on axial gap self bearing motor drives," in *Proc. Int. Symp. Micro/Nano System Technol.*, Hanoi, Vietnam, Dec. 2008, CD-ROM.
- [13] D. Q. Nguyen and S. Ueno, "Sensorless speed control of a permanent magnet type axial gap self bearing motor," in *Proc. 11th Int. Symp. Magn. Bearings*, Nara, Japan, Aug. 2008, CD-Rom.
- [14] D. Q. Nguyen and S. Ueno, "Sensorless speed control of a permanent magnet type axial gap self bearing motor," *J. Syst. Des. Dyn.*, vol. 3, no. 4, pp. 494–505, Jul. 2009.

- [15] A. E. Fitzgerald, C. Kingsley Jr., and S. D. Uman, *Electric Machinery*, 5th ed. New York: McGraw-Hill, 1992.
- [16] A. Chiba, T. Fukao, O. Ichikawa, M. Oshima, M. Takemotor, and D. G. Dorrell, *Magnetic Bearings and Bearingless Drives*, 1st ed. Oxford, U.K.: Elsevier, 2005.



Quang Dich Nguyen (S'09) was born in Bac Ninh, Vietnam. He received the B.Sc. degree in electrical engineering from Hanoi University of Technology, Hanoi, Vietnam, in 1997, and the M.Sc. degree in electrical engineering from Dresden University of Technology, Dresden, Germany, in 2003. He is currently working toward the Ph.D. degree at Ritsumeikan University, Kusatsu, Japan.

In 2000, he joined the Department of Industrial Automation, Hanoi University of Technology. His current research interests include magnetic bearings, self-bearing motors, and sensorless motor control.



Satoshi Ueno (S'99–A'00–M'99) received the B.S.M.E., M.S.M.E., and D.Eng. degrees from Ibaraki University, Hitachi, Japan, in 1995, 1997, and 2000, respectively.

From 2000 to 2004, he was a Research Associate in the Department of Intelligent Machines and System Engineering, Hirosaki University, Hirosaki, Japan. He is currently an Associate Professor of mechanical engineering at Ritsumeikan University, Kusatsu, Japan. His research interests include active magnetic bearings, self-bearing motors, and their applications.

Dr. Ueno is a member of the Japan Society of Mechanical Engineers, Institute of Electric Engineers of Japan, and several other societies.